

README

ATTENTION ENCODER-DECODER NEURAL MACHINE TRANSLATION (ENGLISH TO FRENCH)

■ OVERALL APPROACH

An Attention Encoder-Decoder model that translates from English to French where the encoder is using bidirectional LSTM and the Attention is Luong dot product attention. The word embedding is FastText and the Evaluation metric is BLEU and ROUGE.

1. SELECTED WORD-EMBEDDING METHOD (FASTTEXT)

FastText builds the word embedding by passing the vocab to a fully connected Neural Network where the number of embeddings is the number of activation functions in that neural network and unlike Word2Vec the unit the neural network is trained on is Character N-gram where each word is divided to subwords of N characters and this approach is very useful because it handles Out-Of-Vocabulary because the neural network learns the micro semantic meanings between n-grams.

And also as Word2Vec it has two architectures:

- **Continuous Bag of Words:** Where the surrounding words are used to predict a suitable word for the context.
- **Skip-gram:** Where a word is used to predict the suitable surrounding words to build a context.

2. BIDIRECTIONAL LSTMS

Lets the model go through the sentence from its last word to its first word not only from first to last which helps the model grasp the context more and handle words that can have double meaning.

3. ATTENTION MECHANISM

The attention mechanism takes the hidden states of each LSTM cell in an input step, (each step) in the encoder and compares it to the hidden states of each cell of the decoder's hidden states using dot product similarity and using the results it computes the weighted sum which is called context vector (how much each output of the encoder should affect the prediction).

Then these context vectors are fed alongside with the hidden states of the decoder for the step we are in to a fully connected layer so it predicts the word-vector of the output token.

4. EVALUATION METRICS (BLEU AND ROUGE)

BLEU Metric:

BLEU simply compares the generated text to a reference text through the equation:

$$BP \times \exp(\sum(w \times \log(P)))$$

which has a range of [0, 1].

P is the clipped number of words generated that match occurs in the reference text over the total number of words generated. To take the order of words into consideration for the comparison, we use BLEU2, BLEU3, and BLEU4 for comparing bigger clusters of words that would respect their order.

ROUGE Metric:

ROUGE also compares the generated text to a reference written by a human, where it calculates:

- **Recall:** (no. words generated that match the reference) / no. words in reference
- **Precision:** (no. words generated that match the reference) / no. words generated
- **F1-score:** = $2 \times (\text{precision} \times \text{recall} / (\text{precision} + \text{recall}))$

To take the order into consideration, ROUGE-n uses n-grams to compare n-grams of words that respect the order. Also, LCS (Longest Common Sub-sequence) is used to count the longest chain of words that matches the reference word-wise and order-wise.

5. TRANSLATION RESULTS & BENCHMARK

Qualitative Samples:

Hi	<i>salut</i>
I'm Fares	<i>je suis</i>
I really hope I'm doing well right now	<i>j me espère vraiment que je vais bien en faire</i>
There is still so much to learn	<i>il y a encore tant à apprendre</i>
Am I good enough?	<i>suis je assez bien</i>
Well I join?	<i>bien je me</i>
Should I accept my fate, and give up?	<i>dois je accepter mon destin et un</i>

Comparative Translation Evaluation Table:

#	English	Reference (FR)	Predicted (FR)
0	he made up a story about the dog	il inventa une histoire au sujet du chien	il a inventé une histoire au sujet du chien
1	argued with him about it	je me suis disputé avec lui à ce propos	j'en suis disputée avec lui

#	English	Reference (FR)	Predicted (FR)
2	there is nothing like cold beer on a hot day	il n y a rien de tel qu une bière fraîche un jour de canicule	il n y a rien de frais de prendre de la bière par jour chaud
3	i know the reason but i cannot tell you	je connais la raison mais je ne peux pas vous la dire	je connais la raison mais je ne peux pas te le dire
4	my children do not listen to me	mes enfants ne m écoutent pas	mes enfants refusent de m écouter
5	I got dumped	je ai été larguée	je ai été largué
6	can you determine what happened	peux tu déterminer ce qui s est passé	pouvez vous déterminer ce qui s est passé
7	i wish him the very best in his future endeavors	je lui souhaite le meilleur dans ses futures entreprises	je lui souhaite le meilleur dans ses futures entreprises
8	tom is always watching tv	tom regarde toujours la télévision	tom regarde tout le temps la télé
9	i will leave it up to you	je vous laisse la décision	je vous laisse en décider
10	I know you want to make me happy	je sais que tu veux me rendre heureuse	je sais que vous voulez me rendre heureux
11	tom s coming	tom vient	tom vient

Test-Set Translation Quality Chart:

